

10.7.26

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October 3, 2025

Question

- ① The focal chord to $y^2 = 16x$ is tangent to $(x - 6)^2 + y^2 = 2$, then the possible values of the slope of this chord are: (2003)
- ① $-1, 1$
 - ② $-2, 2$
 - ③ $-2, -\frac{1}{2}$
 - ④ $2, -\frac{1}{2}$

Theoretical Solution

Let us solve the given equation theoretically and then verify the solution computationally

According to the question,

The parabola is

$$y^2 = 16x. \quad (1)$$

In quadratic form, this can be written as

$$\mathbf{x}^T \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \mathbf{x} + 2 \begin{pmatrix} -8 \\ 0 \end{pmatrix}^T \mathbf{x} = 0. \quad (2)$$

The circle is

$$(x - 6)^2 + y^2 = 2 \Rightarrow x^2 + y^2 - 12x + 34 = 0, \quad (3)$$

which can be expressed as

$$\mathbf{x}^T \mathbf{I} \mathbf{x} + 2 \begin{pmatrix} -6 \\ 0 \end{pmatrix}^T \mathbf{x} + 34 = 0. \quad (4)$$

Theoretical Solution

Thus,

$$\mathbf{V} = \mathbf{I}, \quad \mathbf{u} = \begin{pmatrix} -6 \\ 0 \end{pmatrix}, \quad f = 34. \quad (5)$$

The focus of the parabola is

$$\mathbf{F} = \begin{pmatrix} 4 \\ 0 \end{pmatrix}. \quad (6)$$

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Any line through $\mathbf{F}$ with slope m can be written as

$$y = m(x - 4). \quad (7)$$

Theoretical Solution

In normal form, this becomes

$$\mathbf{n}^T \mathbf{x} = c, \quad (8)$$

where

$$\mathbf{n} = \begin{pmatrix} -m \\ 1 \end{pmatrix}, \quad c = \mathbf{n}^T \mathbf{F} = -4m. \quad (9)$$

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For a conic $\mathbf{x}^T \mathbf{V} \mathbf{x} + 2\mathbf{u}^T \mathbf{x} + f = 0$, the line $\mathbf{n}^T \mathbf{x} = c$ is tangent if

Theoretical Solution

$$\left(\mathbf{n}^T \mathbf{C} - c\right)^2 = \left(\mathbf{n}^T \mathbf{V}^{-1} \mathbf{n}\right) \left(\mathbf{u}^T \mathbf{V}^{-1} \mathbf{u} - f\right), \quad (10)$$

where $\mathbf{C} = -\mathbf{V}^{-1} \mathbf{u}$ is the centre.

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Here,

$$\mathbf{C} = -\mathbf{V}^{-1} \mathbf{u} = \begin{pmatrix} 6 \\ 0 \end{pmatrix}, \quad \mathbf{u}^T \mathbf{V}^{-1} \mathbf{u} - f = 36 - 34 = 2. \quad (11)$$

So,

$$\left(\mathbf{n}^T \mathbf{C} - c\right)^2 = \left(\mathbf{n}^T \mathbf{n}\right)(2). \quad (12)$$

Theoretical Solution

Now,

$$\mathbf{n}^T \mathbf{C} = \begin{pmatrix} -m & 1 \end{pmatrix} \begin{pmatrix} 6 \\ 0 \end{pmatrix} = -6m, \quad (13)$$

and $c = -4m$. Thus,

$$(-6m - (-4m))^2 = 2(m^2 + 1). \quad (14)$$

$$(-2m)^2 = 2(m^2 + 1). \quad (15)$$

$$m = \pm 1. \quad (16)$$

Hence, the slopes of the focal chord are

$$\boxed{-1 \text{ and } 1}. \quad (17)$$

Plot

